

Generative AI Foundation Models



Estimated time needed: 45 minutes

Welcome to the hands-on lab generative AI foundation models!

What are generative AI foundation models ?

Models with capabilities like natural language processing and computer vision must be built completely from scratch, requiring resources in terms of data, computing, and processing throughout the whole machine learning cycle.

Foundation models are pre-trained models trained on vast amounts of data. They can be fine-tuned for specific tasks such as generating text, images, audio, and videos based on specific use cases in generative AI. Using deep learning algorithms, these models are highly accurate at recognizing patterns and making predictions.

In this lab, you will use foundation models for question-answer and text classification. You will first focus on classifying text and detecting sentiment using Google's `flan-ul2` model. You will then use `gpt-4o-mini`, which is a large-scale transformer-based foundation model, to get answers to your questions. Finally, you will also review various foundation models and their distinguishing features.

Learning Objectives:

1. Describe the foundation models of generative AI
2. Use pre-trained generative AI foundation models for downstream tasks
3. Classify text and detect its sentiment using the **google/flan-ul2** model
4. Get answers to your questions using **GPT-5 Nano** model
5. Describe the distinguishing features of various foundation models

Generative AI Classroom

Note: Please refer to the section 'Know your generative AI classroom lab' to understand the Generative AI classroom's user interface. If you are already familiar with the interface and its elements, you can skip this section and navigate to the next section to start Exercise 1.

Know your generative AI classroom lab

You will use the generative AI classroom as a prompting tool to learn more about Prompt Engineering. As you open this lab, you are likely to notice something different. You will have the generative AI classroom environment and instructions on one page in a browser. The instructions will be on the left half of the screen, and the generative AI classroom will be on the right half of the screen.

Notes:

1. The prompts used in this lab are for your reference only. You can create your own prompts and generate responses using generative AI.
2. Since AI-generated outputs are dynamic, you may receive different responses even though you've used the same prompt from this lab.

The image shows two side-by-side screenshots. The left screenshot is a document titled "Generative AI Foundation Models" from the Skills Network. It includes a welcome message, a definition of generative AI foundation models, and a list of learning objectives. The right screenshot is the "untitled chat" interface of the AI Classroom. It features a model selector set to "meta-llama/llama-3-2-1b-instruct", a "PROMPT INSTRUCTIONS" section, a "How to use AI Classroom" guide with four steps, and a "Start chat" button. The interface is powered by watsonx.

Generative AI Foundation Models

Skills Network

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How to use AI Classroom

- 1 Name the chat
- 2 Choose your model(s). You may select multiple models when "Compare Models" toggle is active
- 3 Set the prompt instructions
- 4 Chat with your language model

Want to elevate your AI learning experience? Check out [IBM watsonx.ai](#) now!

Type your message

powered by **watsonx**

The generative AI classroom tool allows you to write and compare your prompts to generate desired text with real-time chat responses. You can interact with the language model using the message and chat fields. Moreover, you can choose from multiple generative AI models and learn about their strengths and weaknesses.

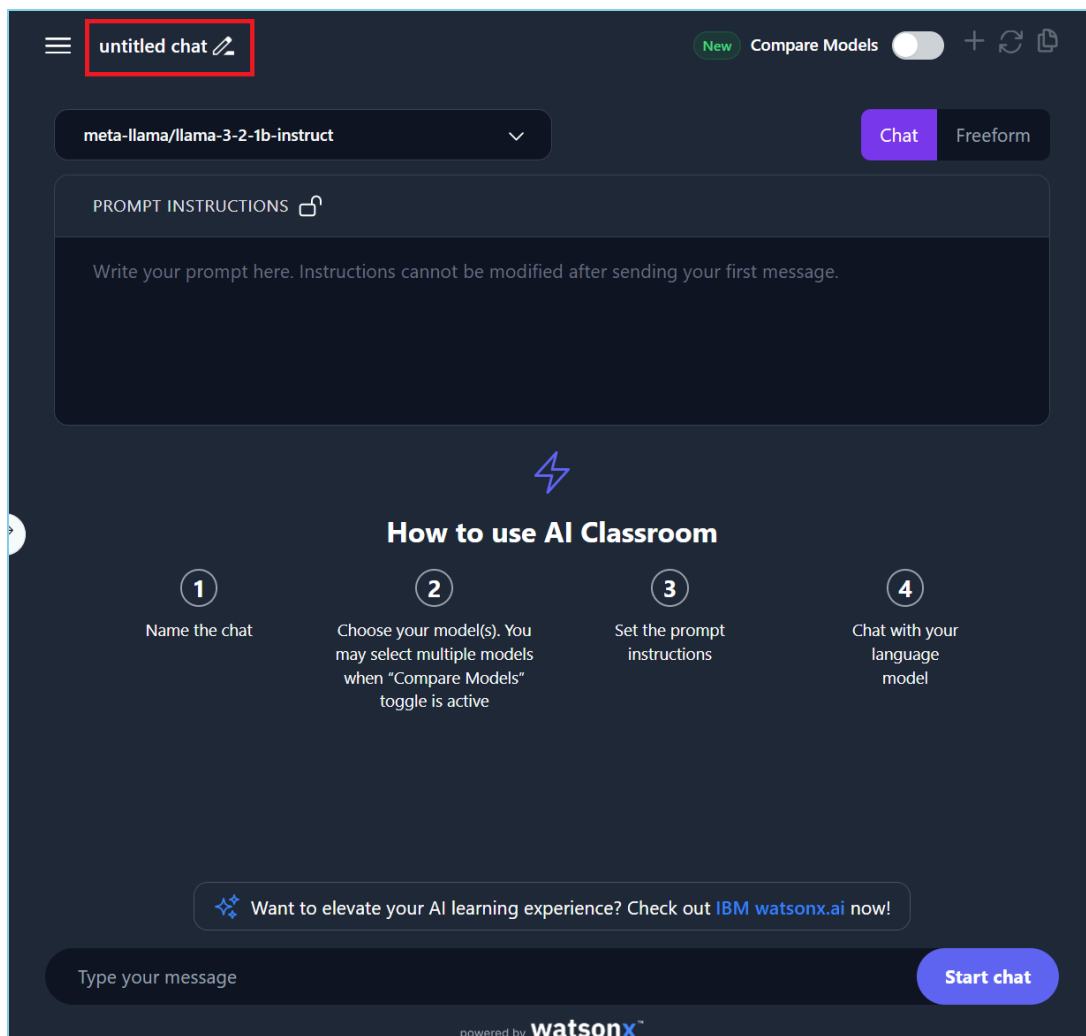
Let's learn about the AI classroom in greater detail.

Step 1: Set up the AI classroom

As a first step, you must set up your AI classroom for better learning experience.

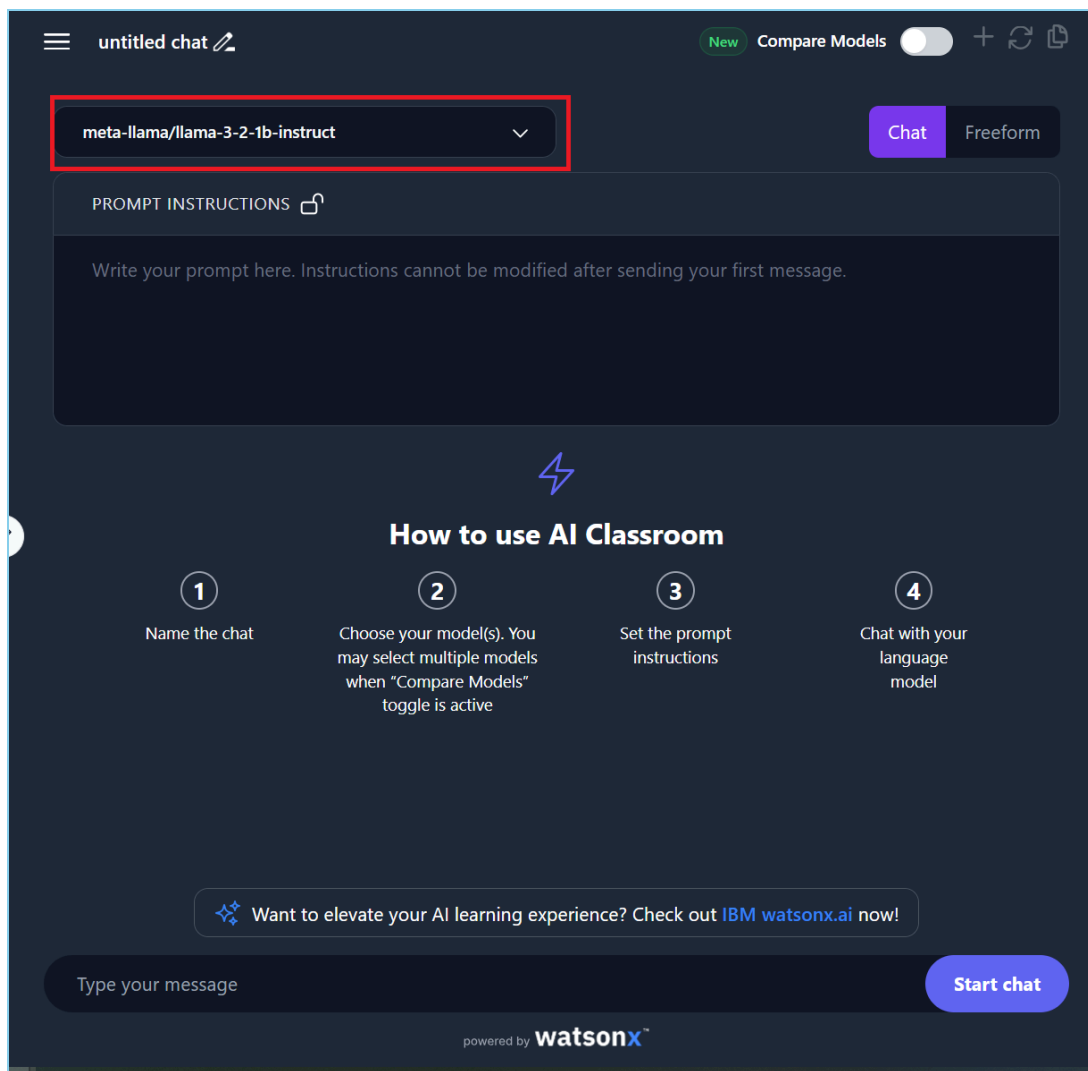
Before beginning with the exercises, you must know the following:

1. **Name the chat:** Use the pencil icon available on the upper-left corner of the right pane to name the **Untitled chat**.



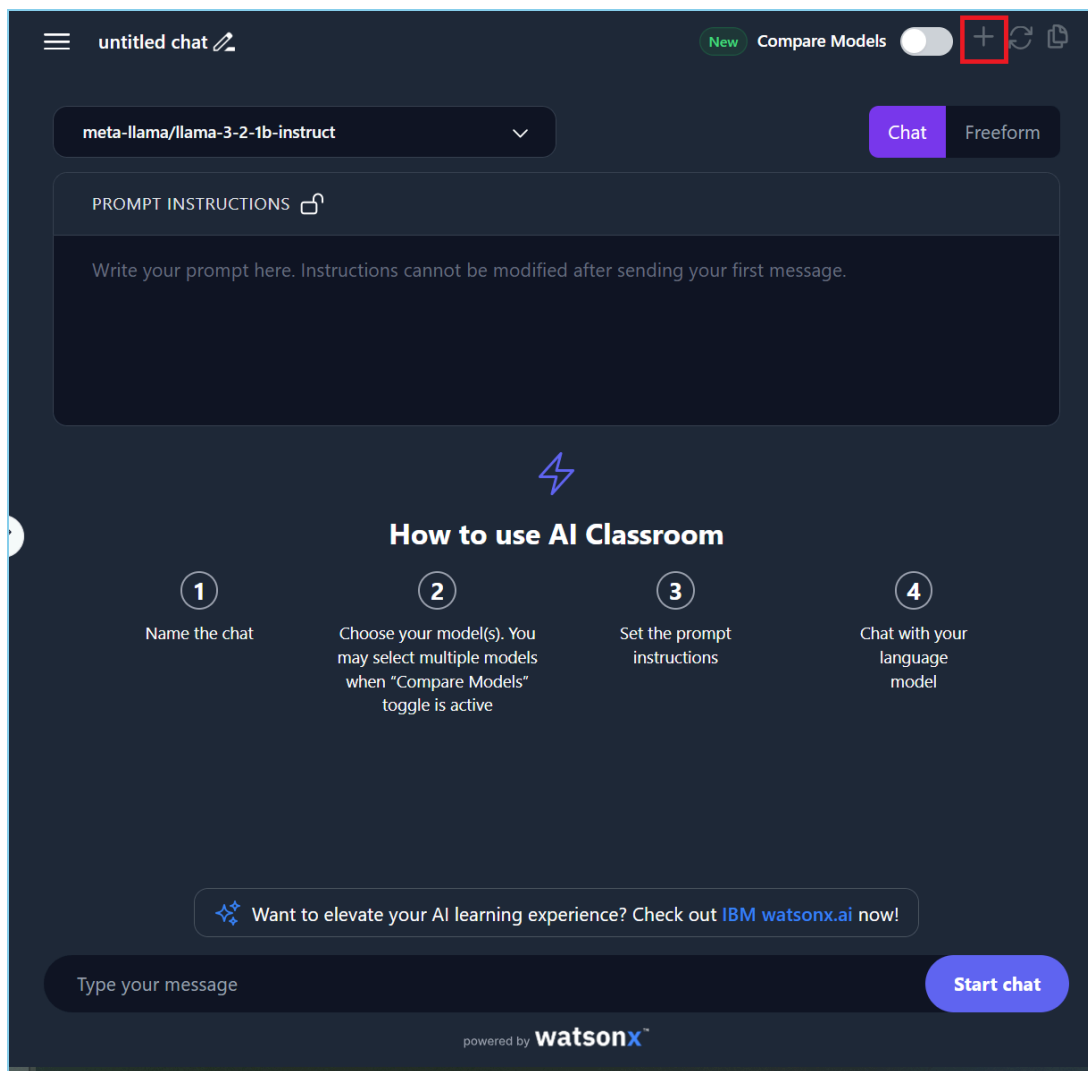
You will find it useful to be able to quickly recognize chat sessions once you start to accumulate many of them in your history. You can find the list in the hamburger menu to the left of **untitled chat**.

2. **Choose the model:** Use the drop-down option from the upper-right corner of the right pane to select the foundation model for the exercise.

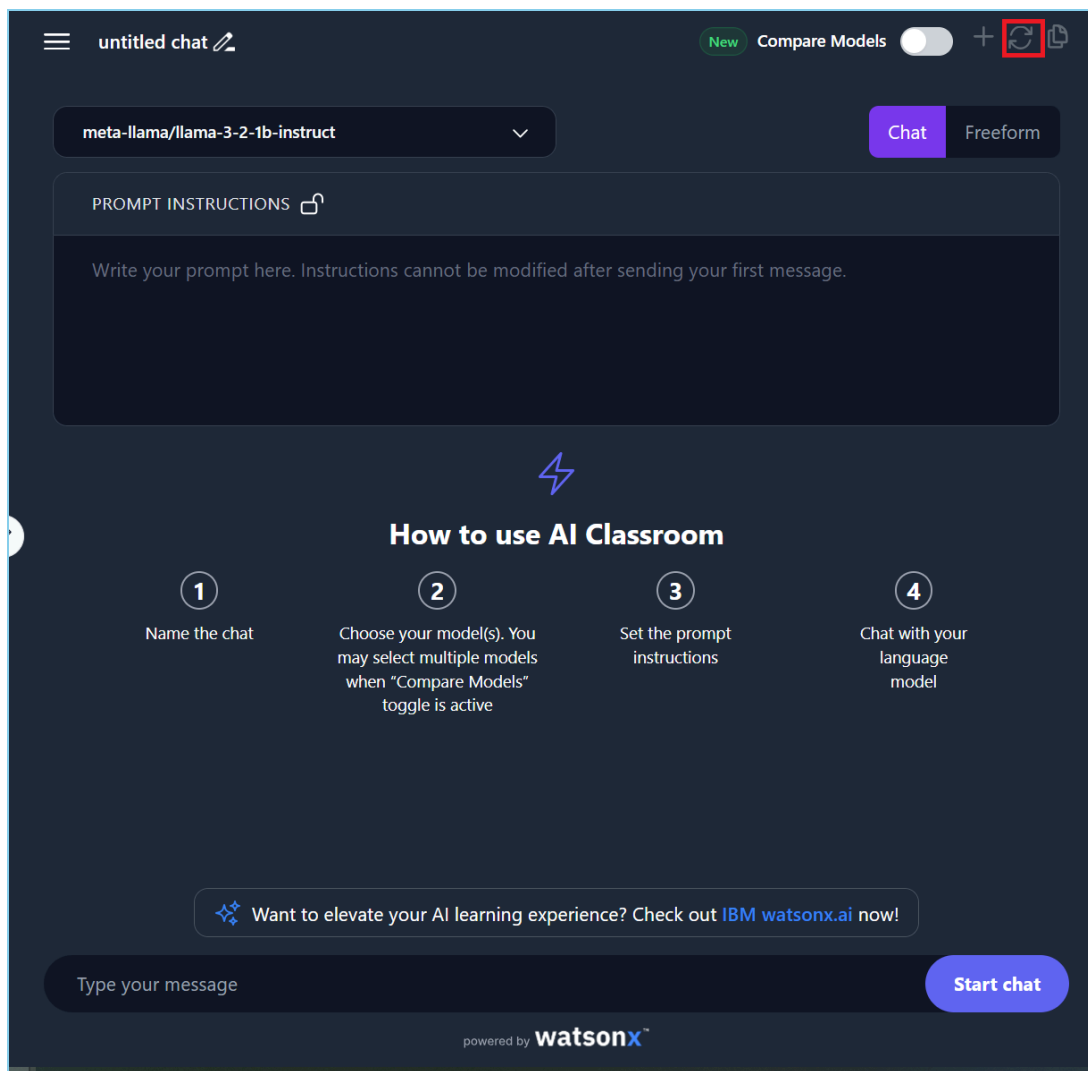


The drop-down option defaults to the gpt-3.5-turbo model; however, you can select alternative large language models from the drop-down list.

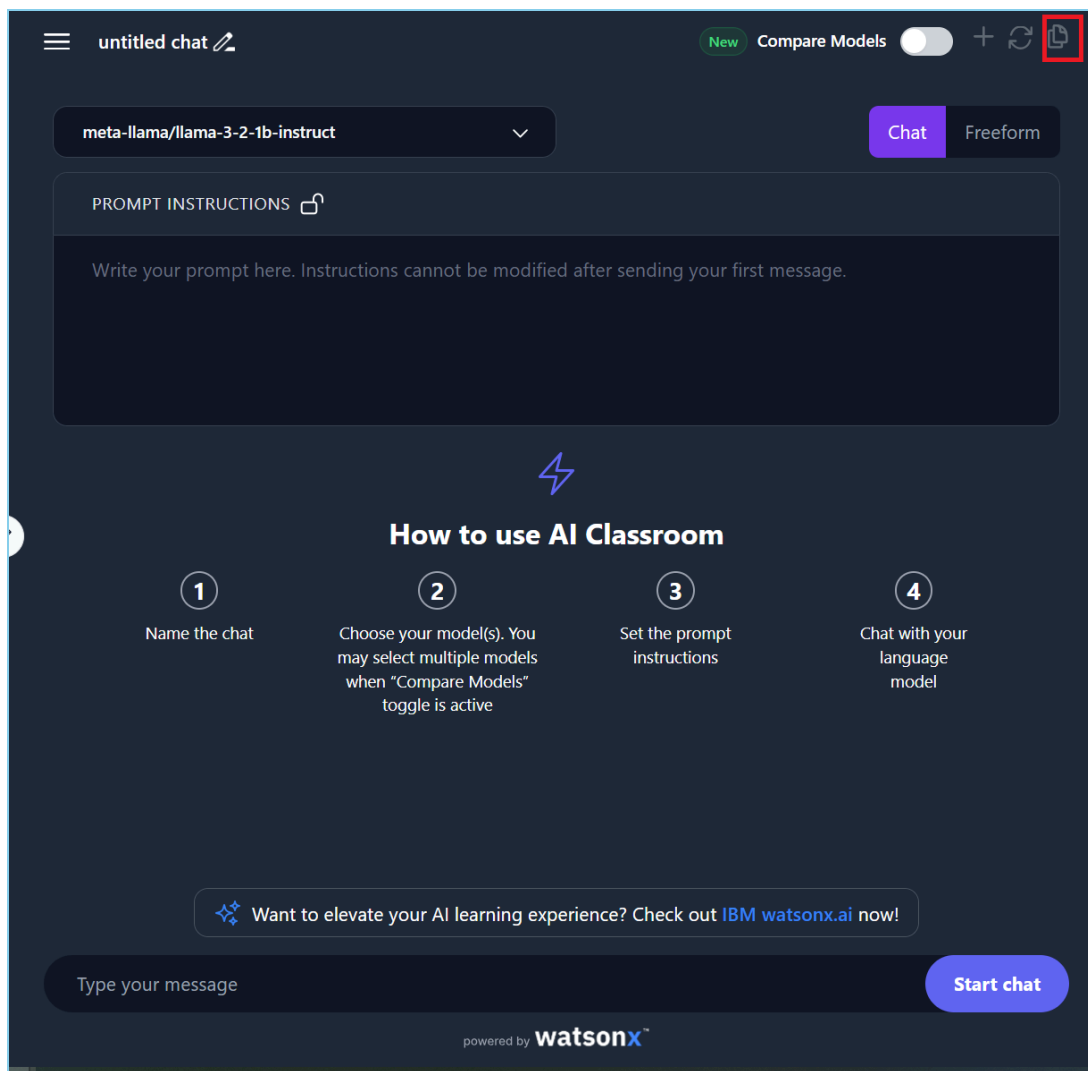
3. **Manage your chats:** Using the three icons available in the upper right corner of the right pane, you can
Create a new chat using the **New Chat** option to start a fresh conversation.



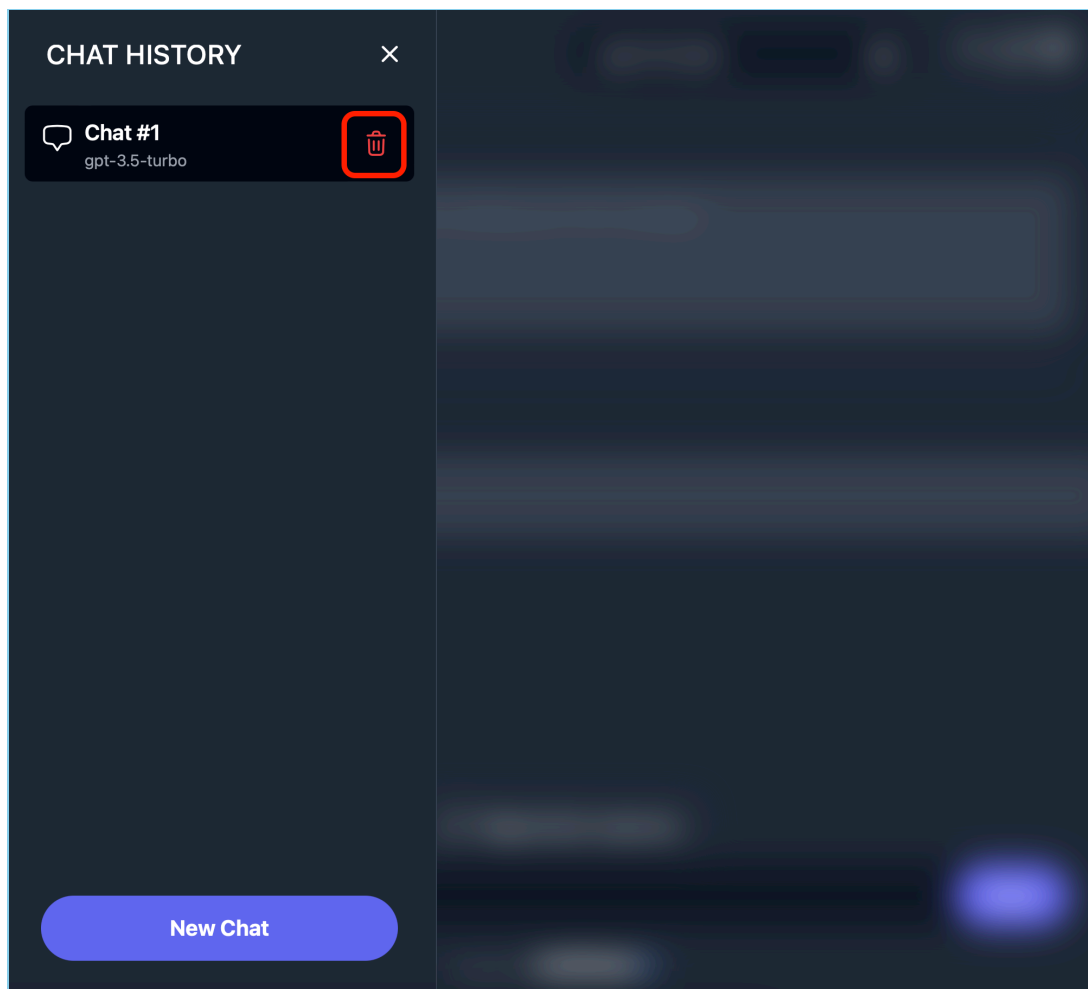
Reset Chat if you want to remove all the questions and answers from the page for the locked prompt instructions.



Duplicate Chat to modify the prompt instructions and keep a different version of the same context.



Delete a chat using the hamburger menu in the upper left corner of the right pane, selecting the trash bin icon against it, and confirming the deletion.

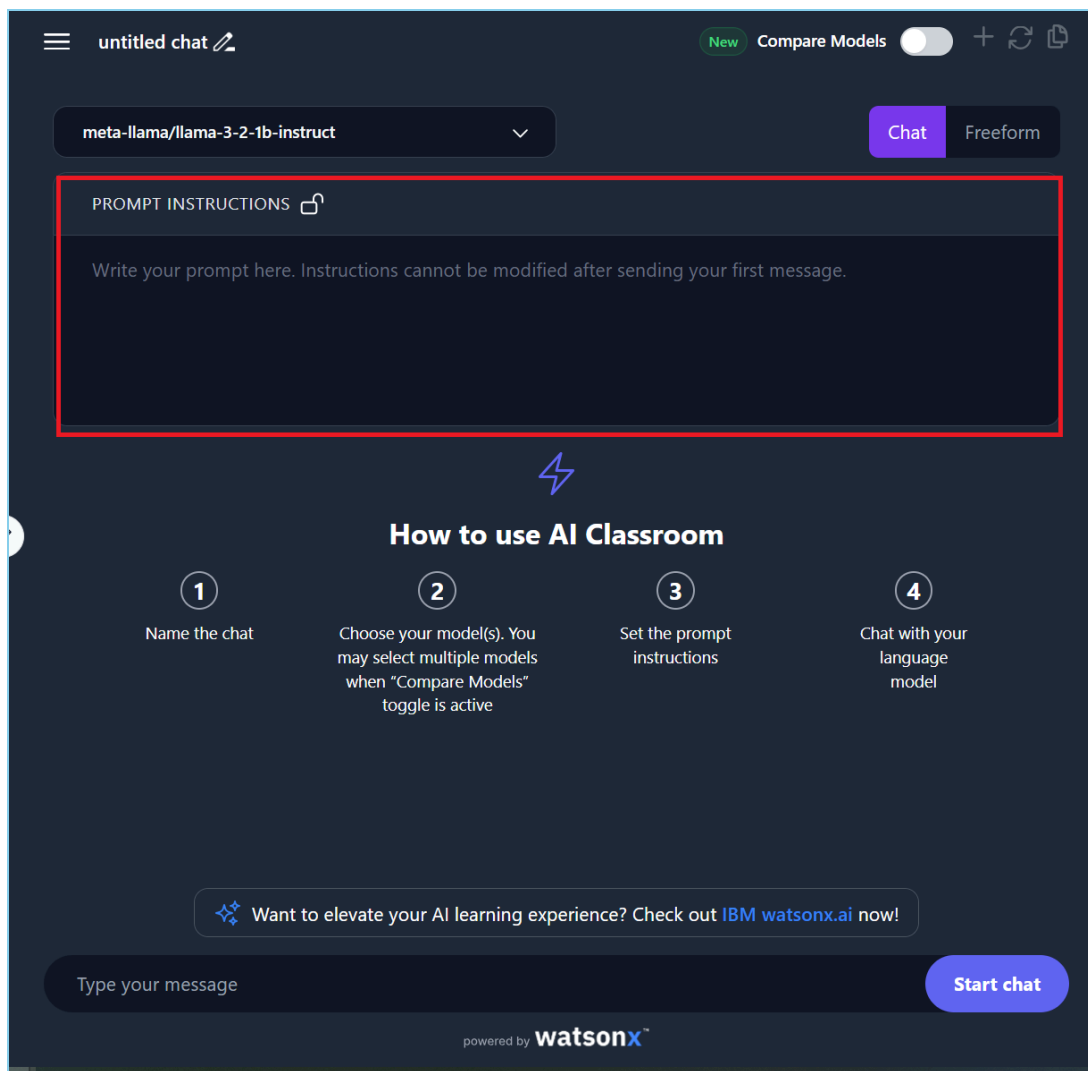


Step 2: Before you begin prompting

You will find two input areas in the right pane: One area in the middle part for your prompt instructions and another area in the lower part for your actual prompts and questions.

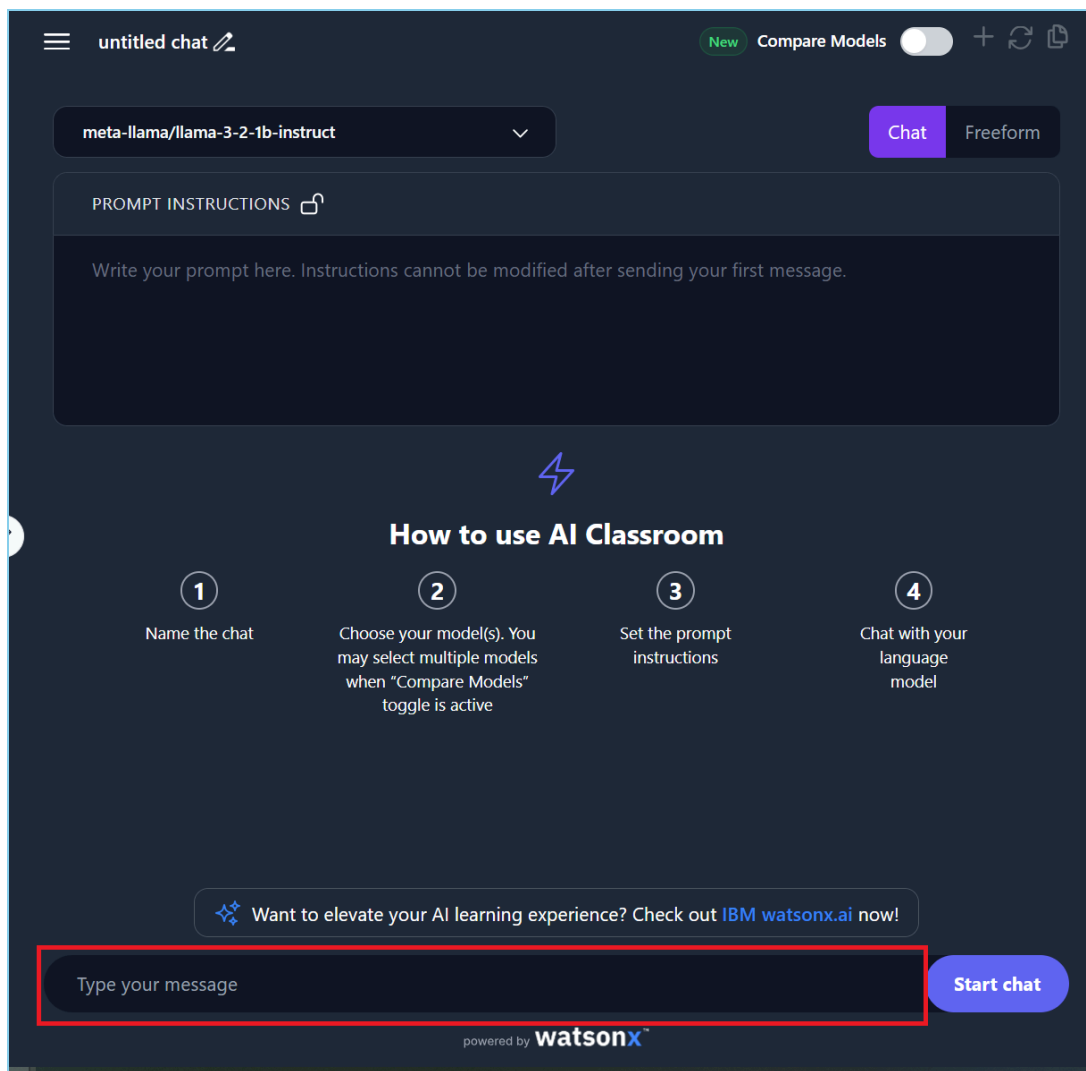
To prompt the foundation model-based chatbot, you must know the following:

1. **Give prompt instructions:** Use the **Prompt Instructions** field on the upper right pane of the chat system to provide instructions or any specific details about the context of the required output.



The prompt instructions allow you to instruct the AI about the context of the conversation in a specific manner. These instructions get locked when you start the chat and cannot be modified later.

2. **Type your message:** Use the text box at the bottom of the page to write the prompts and converse with the chat system. You can use this field to dive into the conversation by sending additional questions or points.



Now, you can experiment with the prompt instructions by providing central context and trying a few queries.

For example, try Talk to me like I am a 5-year-old and then ask difficult questions like How does Artificial Intelligence work?.

Exercise 1: Classify text and detect sentiment using a generative AI foundation model

The generative AI foundation models excel in language understanding tasks such as text classification and sentiment analysis, among others. Their system operates by connecting each output element to each input element and calculating weights between them dynamically according to their connections.

In this exercise, you will learn about text classification using the foundation model **google/flan-ul2**, a powerful natural language processing (NLP) model, and detect the sentiment and tone of your text.

About large language model google/flan-ul2

Google's flan-ul2 foundation model is an encoder-decoder model based on the T5 architecture. It is a 20-billion-parameter model trained on a massive data set of text and code. When generating text or answering questions, Google AI's large language model can consider up to 4096 tokens of context.

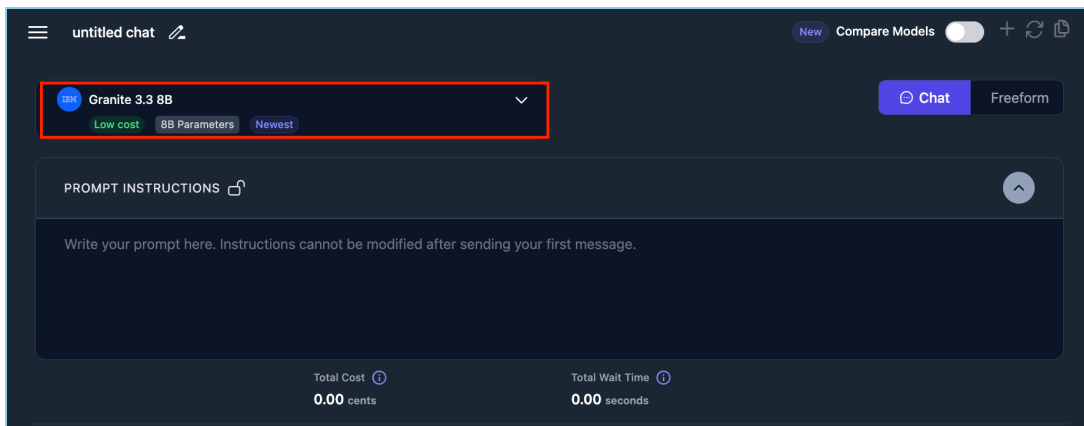
This foundation model can help you generate text such as poems, code, and scripts. It can assist you in translating languages, creating creative content, and providing informative answers to your questions.

It can be commercially used for various applications like language generation (with automated and human evaluation), language comprehension, text classification, question answering, common sense reasoning, long text reasoning, structured knowledge grounding, and data retrieval.

Let's begin experimenting with this large language model.

Step 1: Set your AI environment and select the foundation model

1. As a first step, you can name the chat per your context.
2. Select the large language model **ibm/granite-3-3-8b-instruct** in the upper section of the right pane for this exercise.

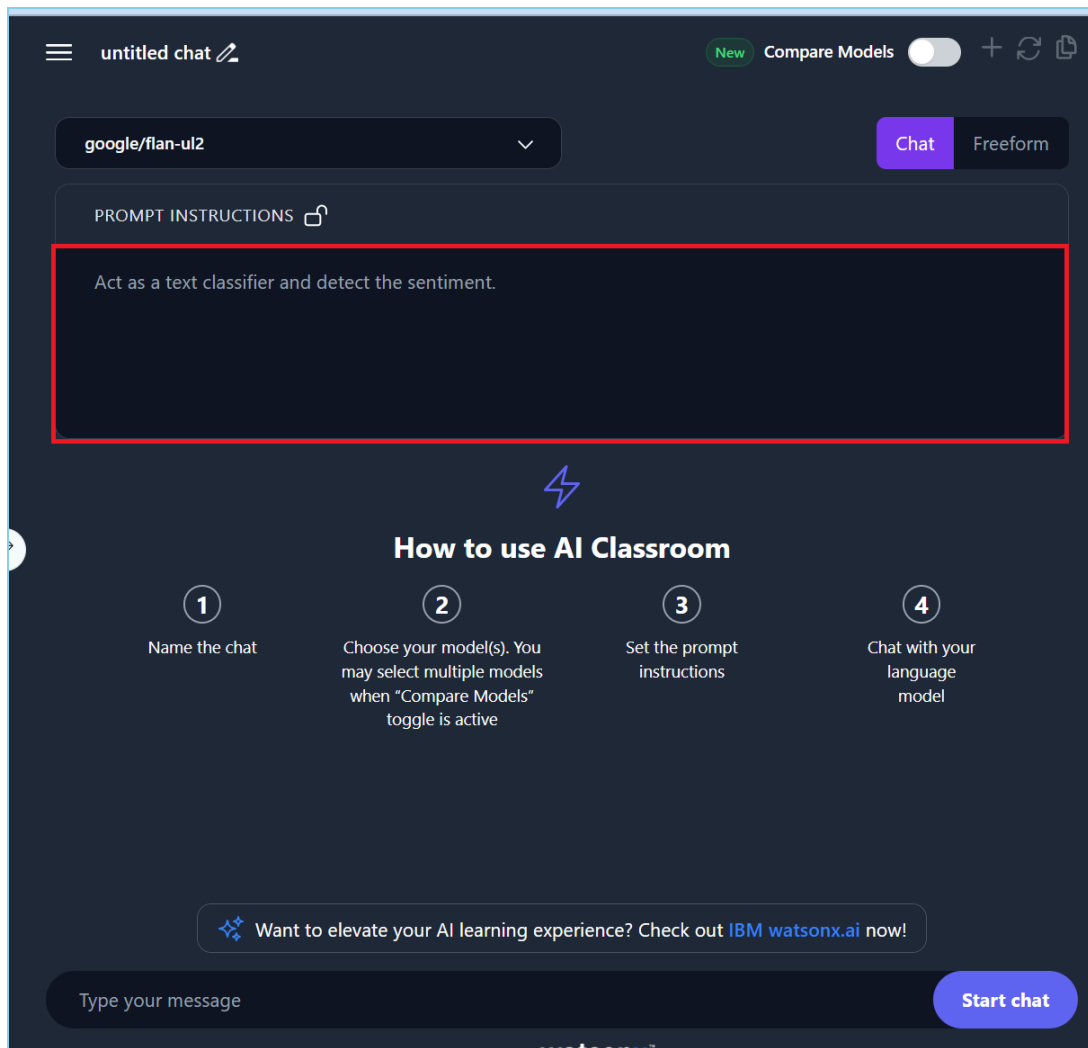


Step 2: Write a relevant prompt

1. Consider a context for which you want to classify text and detect sentiment.
2. Provide relevant instructions in the **Prompt Instructions** field.
For example,

Act as a text classifier and detect the sentiment.

You can copy the above example and paste it in the required field.

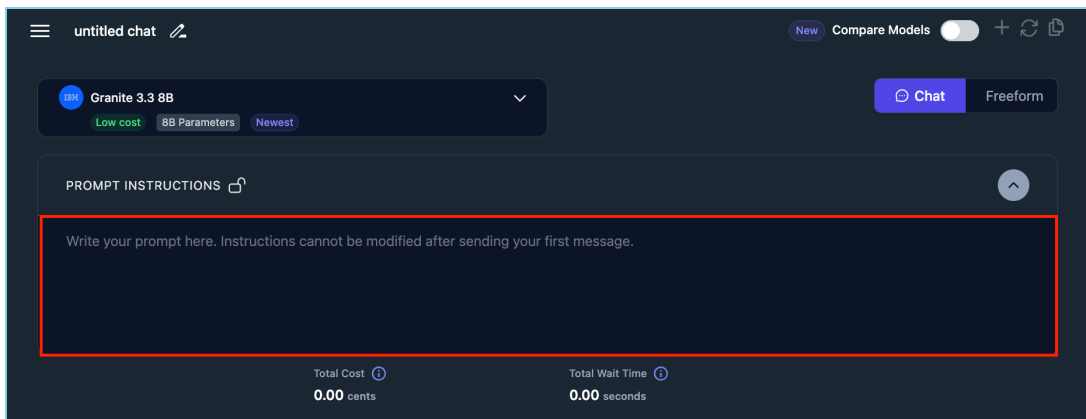


Note: You can include more information or instructions central to your context in the **Prompt Instructions** field.

3. Write the prompt in **Type your message** field.
For example,

My least preferred fruit is mango.

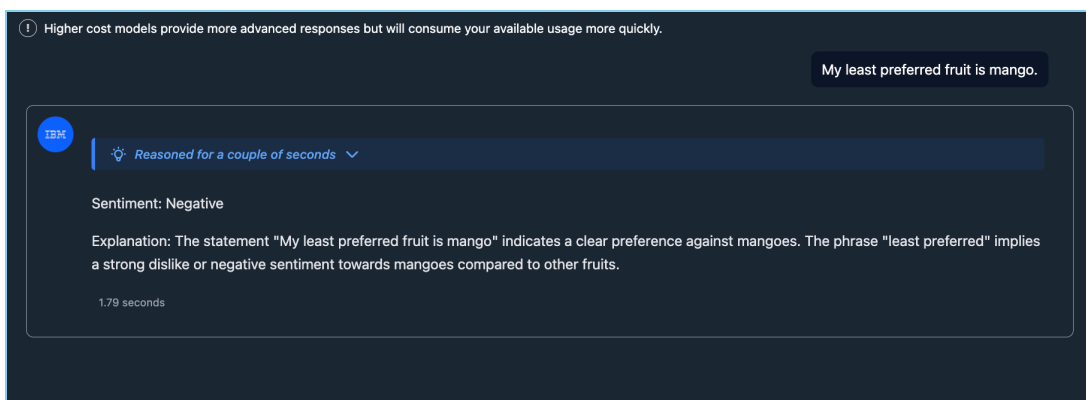
You can copy the above example and paste it in the required field.



Note: You can include additional information or instructions about the context of the required output and your prompt in the **type your message** field.

- Click **Start Chat** to get the chatbot response in the middle part of the right pane. You will notice the response similar to the following:

Sentiment: Negative



- Let's try to test another sentiment. Write another prompt in **Type your message** field. For example,

Yesterday's papaya was delicious.

You can copy the above example and paste it in the required field.

6. Click **Start Chat** and review the chatbot response like the following:

Sentiment: Positive



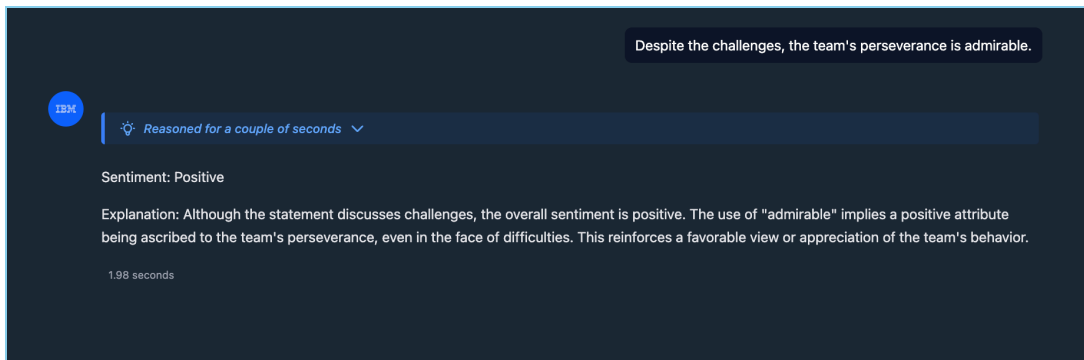
7. Let's try another example to test a complex sentiment. Write another prompt in **Type your message** field. For example,

Despite the challenges, the team's perseverance is admirable.

You can copy the above example and paste it in the required field.

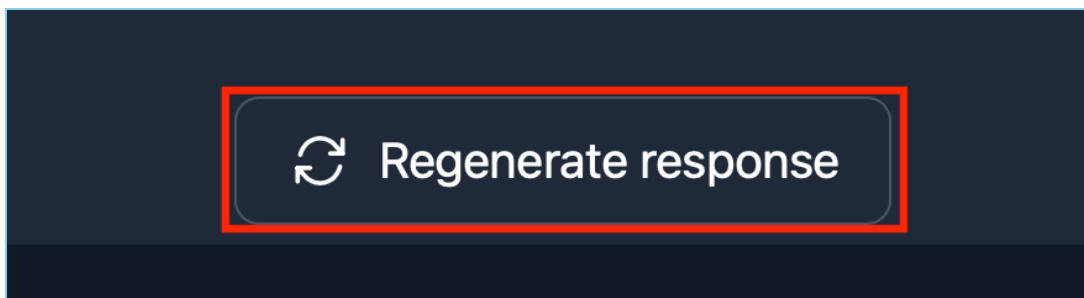
8. Click **Start Chat** and review the chatbot response like the following:

Sentiment: Positive



Step 3: Review results and responses

1. Continue interacting with the chatbot to test more sentiments.
For example,
 - I am so glad to see you today.
 - The tech conference in Berlin was amazing.
 - I feel sad to see any biker fall in the motor race.
 - The restaurant has a good rating, but the food is below average.
2. Use the **Regenerate response** option to get another version of the responses to your latest question.



Note: Observe how the foundation model **google/flan-ul2** helps classify the text and detects its sentiment and tone. This capability can be used to develop apps that can perform sentiment analysis and improve content quality.

Exercise 2: Get answers to your questions using the foundational model

Given their excellent language understanding, the foundation models help with natural language tasks, such as question-answer solutions, among others.

In this exercise, you will find answers to your questions using the **gpt-4o-mini** foundation model.

About language model gpt-4o-mini

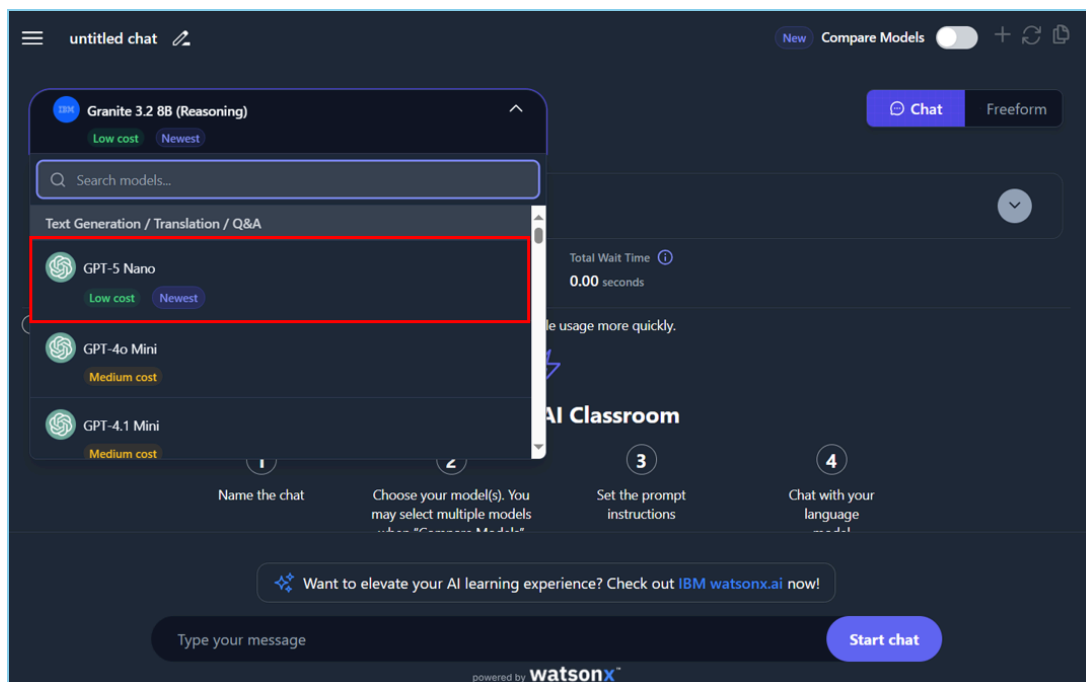
OpenAI's gpt-4o-mini is an advanced large language model built on a refined decoder-only transformer architecture. Designed for enhanced natural language understanding and generation, gpt-4o-mini offers improved contextual awareness, coherence, and accuracy compared to its predecessors. This model is engineered for a broader range of tasks with greater performance and efficiency.

GPT-4o-mini utilizes a more optimized training process and advanced algorithms, likely surpassing the capabilities of the 175-billion-parameter GPT-3.5-turbo model, though the exact parameter count is undisclosed. Its vast capabilities allow it to generate high-quality text and perform various tasks such as content creation, conversational AI, language translation, code generation, and more.

Key applications for gpt-4o-mini include chatbots, virtual assistants, automated content generation, and advanced problem-solving across industries. It can handle nuanced and complex tasks, such as answering intricate queries, writing creative content, generating technical scripts, and assisting in educational and professional settings..

Step 1: Set your AI environment and select the foundation model

1. As a first step, you can name the chat per your context.
2. Select the large language model **GPT-5 Nano** for this exercise.



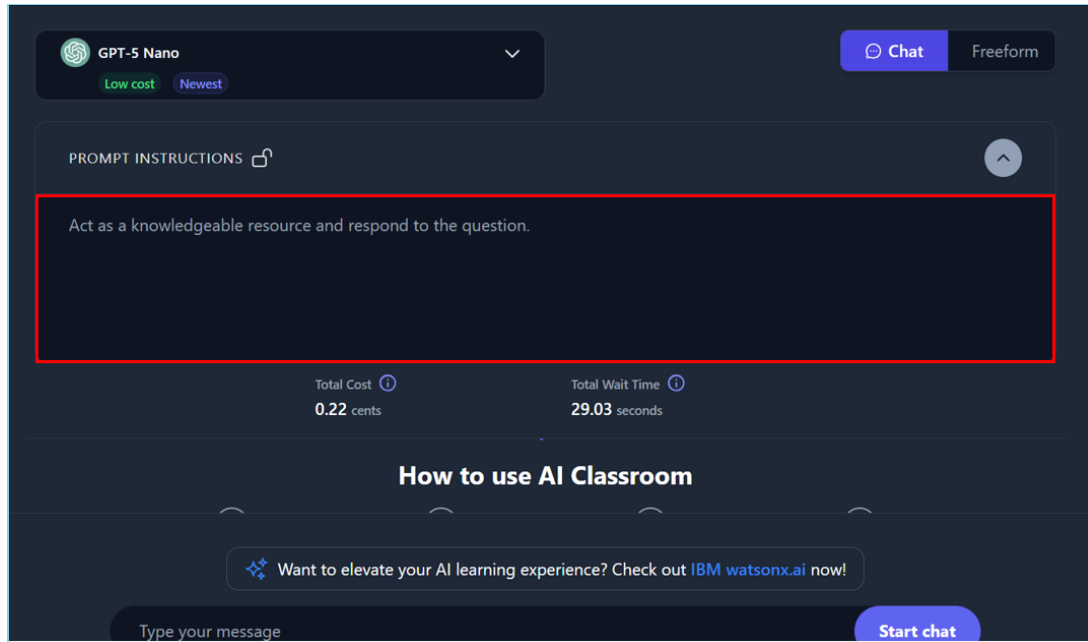
Note: Though the drop-down list defaults to the **gpt-3.5-turbo** model, you can select an alternative foundation model for the exercise.

Step 2: Write a relevant prompt

1. Consider the context of any questions or problems you are trying to solve.
2. Provide the instructions in the **Prompt Instructions** field.
For example,

Act as a knowledgeable resource and respond to the question.

You can copy the above example and paste it in the required field.



Note: You can include more information or instructions central to your context in the **Prompt Instructions** field. Alternatively, you can ignore or skip the prompt instructions completely if you want answers to your questions without any central context.

3. Write the prompt in **Type your message** field.

For example,

What is the longest word in English?

You can copy the above example and paste it in the required field.

Regenerate response

What is the longest word in English?

Send

Note: You can include additional information or instructions with your prompt in the **Type your message** field.

- Click **Start Chat** to get the chatbot response as a result in the middle part of the right pane. You will notice the response similar to the following:

There isn't a single universally accepted longest word in English. The answer depends on criteria such as dictionary inclusion, whether you count coined terms, or whether you treat extremely long scientific or chemical names as words. The word most often cited as the longest in major English dictionaries is pneumonoultramicroscopicsilicovolcanoconiosis, a 45-letter term for a lung disease caused by inhaling very fine silica dust.

Overview

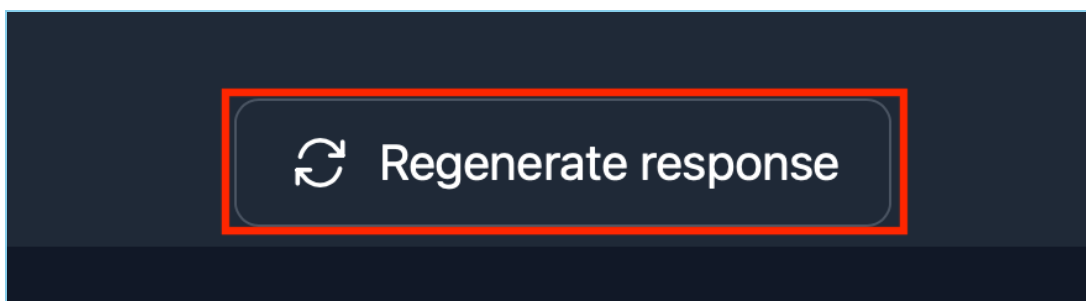
Commonly cited longest English word: pneumonoultramicroscopicsilicovolcanoconiosis (45 letters). It originated as a coined term and is included in several dictionaries. It is not used in everyday language, but it is widely referenced in discussions about word length.

Notable long words

- pneumonoultramicroscopicsilicovolcanoconiosis — 45 letters; a lung disease caused by inhaling silica dust
- pseudopseudohypoparathyroidism — 30 letters; a medical condition name
- floccinaucinihilipilification — 29 letters; the act of describing something as worthless
- antisestablishmentarianism — 28 letters; a historical political movement

Step 3: Review results and responses

- Continue interacting with the chatbot to generate more and more responses.
- Use the **Regenerate response** option to get another version of the responses to your latest question.



Step 4: Ask more questions

- Continue with the same chat window, or create a new chat using the hamburger icon and **New chat** option on the left side of the right pane.
- Consider the context of any questions or problems you are trying to solve.
- If you continue the same chat, you cannot change the **Prompt Instructions** field because it is locked. If you are opening a new chat, you can provide **Prompt Instructions** if required.

For example,

Act as a knowledgeable resource and respond to the question.

You can copy the above example and paste it in the required field.

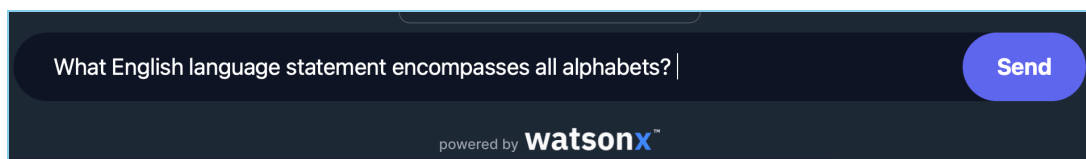
Note: You can include more information or instructions that are central to your context in the **Prompt Instructions** field. Alternatively, you can ignore or skip the prompt instructions completely if your questions have no central context.

4. Write the prompt in **Type your message** field.

For example,

What English language statement encompasses all alphabets?

You can copy the above example and paste it in the required field.



The screenshot shows a dark-themed chat interface. At the top, there is a light blue header bar. Below it, a dark grey input field contains the text "What English language statement encompasses all alphabets? |". To the right of the input field is a blue rounded button with the word "Send" in white. Below the input field, the text "powered by watsonx™" is displayed in a smaller font.

Note: You can include additional information or instructions with your prompt in the **Type your message** field.

5. Click **Start Chat** to get the chatbot response in the middle part of the right pane.
You will notice the response similar to the following:

In English, a sentence that uses every letter of the alphabet at least once is called a pangram. Pangrams are often used for testing fonts, showing font appearance, or practicing typing because they exercise all letters. The easiest-to-remember pangram is: 'The quick brown fox jumps over the lazy dog.' This sentence uses all 26 letters (ignoring case and punctuation).

Examples of pangrams

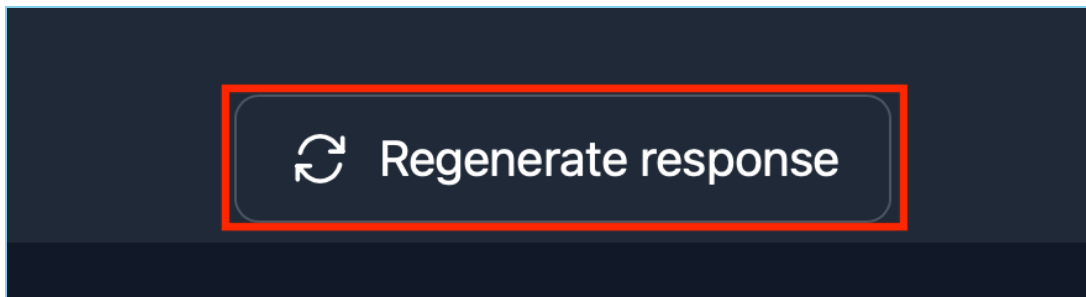
- The quick brown fox jumps over the lazy dog.
- Pack my box with five dozen liquor jugs.
- Sphinx of black quartz, judge my vow.
- How quickly daft jumping zebras vex.

Short pangrams

Some very short pangrams that still cover all letters include: 'Cwm fjord bank glyphs vext quiz.' (a classic 26-letter pangram) and 'Sphinx of black quartz, judge my vow.'

Step 5: Review results and responses

1. Continue interacting with the chatbot to generate more responses.
2. Use the **Regenerate response** option to get another version of the responses to your latest question.



Note: Observe how the foundation model **gpt-4o-mini** helps generate text and respond to questions. This capability can be used to develop apps that generate text and scale up content creation.

Know more about other foundation models

Let's dive into other foundational models available in the generative AI classroom.

The IBM Granite models

The Granite foundation models have two variants: Granite.13b.instruct and Granite.13b.chat, where 13b indicates 13 billion parameters. These variants are built on the Decoder architecture, which forms the foundation for large language models (LLMs) to predict the next word in a sequence.

These multi-size foundation models apply generative AI to both language and code. They facilitate various natural language processing (NLP) tasks like content generation and named entity recognition.

The tiuae/falcon-7b-instruct model

The Technology Institute of Innovation (TII) developed a versatile and powerful large language model, Falcon. Its variant, **falcon-7b-instruct**, is a 7-billion-parameter model based on the decoder-only model trained on 1500 billion tokens of the RefinedWeb. Its architecture helps optimize inference, fine-tuning, quantization, and text generation tasks.

As a powerful text generation tool, the **falcon-7b-instruct** foundation model can be used to generate content in the English language. Besides creating informative and engaging content, such as articles, blog posts, social media posts, and research in various fields, this foundation model can help you create chatbots, virtual assistants, and educational tools.

The meta-llama/Llama-2-7b-chat-hf model

Llama is the large language model from Meta AI. As a collection of generative text models, Llama has been pre-trained and fine-tuned with 7-billion to 70-billion-parameters models. This repository for the 7B fine-tuned model has been modified and optimized for dialogue use cases. This auto-regressive language model uses an optimized transformer architecture.

The **meta-llama/Llama-2-7b-chat-hf** is a versatile text generation tool that can be used for various applications that require tasks such as, following instructions and completing requests thoughtfully, answering questions in a comprehensive and informative way, and generating different creative text formats, such as poems, code, scripts, musical pieces, emails, and letters.

Distinguishing features of various foundation models

Let's review what makes these models unique and significant.

Model	Distinguishing features
gpt-4o-mini	A lightweight, optimized version of GPT-4o, designed to provide faster responses at lower cost while maintaining strong reasoning and language understanding capabilities. It is well-suited for tasks requiring efficiency, such as chat applications, quick content generation, and interactive use cases.
gpt-4-turbo	GPT-4-turbo is an advanced large language model trained on a massive dataset, designed to understand language at a deep level. Its sophisticated architecture allows it to generate highly coherent, contextually accurate, and informative responses across a broad range of prompts. With optimized performance, GPT-4-turbo is well-suited for various applications, including natural language understanding, content creation, code generation, translation, and more.
tiuae/falcon-7b-instruct	Falcon-7B-Instruct is a ready-to-use chat-based or instruct model that follows instructions in a prompt and provides a detailed response. It is mostly trained on English data and cannot be generalized appropriately to other languages. Based on the instruct model, it may be ideal for most text generation tasks other than further fine-tuning. Furthermore, as it is trained on a large-scale corpora representative of the web, it will carry the stereotypes and biases commonly encountered online.
meta-llama/Llama-2-7b-chat-hf	Llama is intended for commercial use and research in English. The scope of this model is limited to the English language and does not generate appropriate completion for other language prompts. It can be further adapted to develop tuned models intended for

Model	Distinguishing features
	assistant-like chat and pre-trained models for various natural language generation tasks. However, this is not an instruct model and may further fine-tune.
IBM Granite models	Recognizing the fact that a single model cannot cater to the distinct demands of every business use case, the Granite models, which are specially designed for businesses, are being developed in multiple sizes. These multi-size foundation models apply generative AI to both language and code.
GPT-5 Nano	GPT-5 Nano is a compact, resource-efficient variant of the GPT-5 family. It is designed to deliver fast responses and lower compute costs while retaining essential reasoning and language capabilities of the larger GPT-5 model. This makes it ideal for edge deployments, mobile applications, and scenarios requiring lightweight inference with good accuracy but constrained resources.

Conclusion

Congratulations! You just completed the hands-on lab generative AI foundation models.

In this lab, you used foundation models of generative AI for various natural language processing (NLP) tasks. You used various prompts to experiment with various foundation models using the generative AI classroom lab. You generated various content using foundation models, including text classification, sentiment identification, and question-answering.

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